ONLINE LOST AND FOUND PORTAL

A PROJECT REPORT

Submitted by

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in partial fulfillment for the requirements for the award degree of

Bachelor of Engineering

CSB1332 - DESIGN PROJECT

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

K RAMAKRISHNAN COLLEGE OF TECHNOLOGY (AUTONOMOUS)

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APRIL 2026

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**ONLINE LOST AND FOUND PORTAL**" has been carried out by the students, **SRIRAM S, SURYA P, YUVARAJ P,** The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

Date of Viva Voce: 5.5.2026

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ABSTRACT

The Online Lost and Found Portal is a web-based application designed to simplify and improve the process of reporting and recovering lost items within an organization or community. It allows users to register, log in, and post detailed information about lost or found items, including descriptions, images, categories, and location details. The system provides a centralized platform where users can search, filter, and identify potential matches, thereby increasing the chances of successful recovery. The application is developed using modern web technologies, ensuring a responsive, user-friendly, and accessible interface across devices. It incorporates structured database management for efficient storage and retrieval of item data. Features such as image uploads, advanced search options, and intelligent or AI-based matching enhance the accuracy and reliability of item identification. Furthermore, the system improves communication between users and provides better transparency through organized data handling. By replacing traditional manual methods, the portal reduces time, minimizes errors, and ensures efficient management of lost and found records. Overall, the system offers a scalable, secure, and effective digital solution for handling lost and found items.

Keywords: Online Lost and Found Portal, Artificial Intelligence (AI), Item Matching, Web Application, Search and Filtering, Database Management, Digital Item Tracking.

ACKNOWLEDGEMENT

We thank our **Dr. N. Vasudevan**, Principal, for his valuable suggestions and support during the course of my research work.

We thank **Mr. R. Rajavarman**, Head of the Department, Assistant Professor (Sr. Grade), Department of Computer Science and Engineering, for his valuable suggestions and support during the course of my research work.

We wish to record my deep sense of gratitude and profound thanks to my Guide **Ms. R. Madhumathi**, Assistant Professor, Department of Computer Science and Engineering for her keen interest, inspiring guidance, constant encouragement with my work during all stages, to bring this thesis into fruition.

We are extremely indebted to our project coordinator **Mrs. S. Gayathri**, Assistant Professor, Department of Computer Science and Engineering, for his valuable suggestions and support during the course of my research work.

We also thank the faculty and non-teaching staff members of the Department of CSE, K Ramakrishnan College of Technology(Autonomous), Trichy, for their valuable support throughout the course of my research work.

Finally, we thank our parents, friends and our well wishes for their kind support.

SIGNATURE

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LIST OF ABBREVIATIONS

AI	-	Artificial Intelligence
AR	-	Augmented Reality
VR	-	Virtual Reality
API	-	Application Programming Interface
Gui	-	Graphical User Interface
DB	-	Database
SQL	-	Structured Query Language
CPU	-	Central Processing Unit
ML	-	Machine Learning
CV	-	Computer Vision
SFM	-	Structure from Motion
HTTP	-	Hypertext Transfer Protocol
JSON	-	JavaScript Object Notation
REST	-	Representational State Transfer
PNG	-	Portable Network Graphics
JPEG	-	Joint Photographic Experts Group
IDE	-	Integrated Development Environment
GPU	-	Graphics Processing Unit

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

The process of managing lost and found items in institutions or public places is often handled manually through registers, notice boards, or word of mouth. While these traditional methods may provide some level of assistance, they are inefficient, time-consuming, and often fail to connect the rightful owner with the lost item. There is a lack of a centralized system to record, track, and match lost and found items effectively.

The *Online Lost and Found Portal* is a web-based system designed to streamline and digitize the process of reporting and recovering lost items. The system allows users to register, log in, and post details about lost or found items, including descriptions, images, and location information. By storing this information in a centralized database, the platform enables users to search and identify potential matches easily.

In recent years, the advancement of web technologies, including modern frontend frameworks and backend systems, has made it possible to develop interactive and efficient platforms for real-time data management. The proposed system utilizes these technologies to provide a user-friendly interface and seamless communication between users. Additionally, integration with intelligent search or API-based features can improve the accuracy of matching lost and found items.

Furthermore, the system offers a structured and accessible platform where users can view, update, and manage item reports. This improves transparency and reduces the chances of lost items going unclaimed. By digitizing the lost and found process, the system enhances efficiency, saves time, and increases the likelihood of successfully recovering lost belongings.

1.2 PROBLEM STATEMENT

In many institutions and public environments, the management of lost and found items is still handled using traditional methods such as manual registers, notice boards, or informal communication. These approaches are unorganized, inefficient, and often lead to loss of important information. As a result, individuals who lose their belongings face difficulty in reporting and recovering them, while those who find items have no proper platform to report or return them.

The absence of a centralized and digital system makes it challenging to track lost items and match them with found ones. Important details such as item description, location, and time may be incomplete or misplaced, reducing the chances of successful recovery. Additionally, manual systems do not provide real-time updates or easy search functionality, making the process time-consuming and frustrating for users.

Furthermore, there is a lack of transparency and accessibility in existing methods, as users cannot easily verify or browse reported items. This often leads to unclaimed items and inefficient handling of lost-and-found records. Therefore, there is a need for a reliable, user-friendly, and centralized digital platform that can efficiently manage lost and found items, improve communication between users, and increase the chances of recovering lost belongings.

1.3 APPLICATION DOMAIN

The *Online Lost and Found Portal* falls under the domain of web-based information management systems, which are designed to collect, store, and manage user-generated data efficiently. This application is specifically developed for environments such as educational institutions, campuses, offices, and public places where incidents of lost and found items frequently occur.

The system operates as a service-oriented platform that enables users to report lost items, submit details of found items, and search for possible matches. It provides a centralized database where all item-related information, including descriptions, images, locations, and timestamps, is stored and managed effectively.

In addition, the application incorporates concepts from domains such as user authentication, database management, and intelligent search systems. With features like secure login, personalized dashboards, and advanced filtering options, the system enhances user interaction and data security. The integration of AI-based assistance further improves the efficiency of matching lost and found items.

Overall, the application domain focuses on developing a reliable, scalable, and user-friendly digital platform that improves communication, reduces manual effort, and increases the chances of recovering lost belongings in a structured and efficient manner.

1.3.1 Item Data

In the *Online Lost and Found Portal*, digital records refer to the information collected and stored about lost and found items within the system. Instead of physical evidence, the system manages structured digital data such as item titles, descriptions, images, locations, dates, and user details.

These records act as the primary source of information for identifying and matching lost items with found ones. Each item entry is stored securely in the database and can be accessed, searched, and analyzed by users through the platform.

1.3.2 Item Documentation

Item documentation in the *Online Lost and Found Portal* refers to the process of systematically recording and maintaining detailed information about lost and found items within the system. Unlike traditional manual methods such as registers or notice boards, this system provides a structured digital approach to document item details accurately.

Users can submit comprehensive information about an item, including its title, description, category, color, location, date, and supporting images. This detailed documentation helps in clearly identifying items and improves the chances of matching lost items with found ones. The inclusion of visual proof further enhances the reliability of the information provided.

1.3.3 Item Mismatching and Visualization

In the *Online Lost and Found Portal*, item matching and visualization refer to the process of identifying potential matches between lost and found items using digital data. Instead of reconstructing a physical scene, the system analyzes item details such as descriptions, categories, locations, dates, and images to find similarities between user reports.

The platform provides an organized and interactive interface where users can view item listings, compare details, and identify possible matches. Advanced search and filtering options allow users to narrow down results based on specific criteria, making the process faster and more efficient.

1.4 AIM AND OBJECTIVE

1.4.1 Aim

The aim of the *Online Lost and Found Portal* is to develop a centralized and user-friendly web-based platform that simplifies the process of reporting, tracking, and recovering lost items. The system is designed to replace traditional manual methods with a more efficient digital solution, enabling users to easily submit and search for lost or found items. It also aims to improve communication between users and increase the chances of successful item recovery through structured data management and intelligent search features.

1.4.2 Objectives

- To design and develop a secure web-based platform for managing lost and found items.
- To provide user authentication features such as registration and login for secure access.
- To enable users to report lost and found items with detailed information including images and location.

- To implement an efficient search and filtering mechanism for quick item identification.

- To provide a user dashboard for managing and tracking posted items.

- To improve the accuracy of matching lost and found items using intelligent or AI-based assistance.

- To ensure data is stored securely and can be easily retrieved when needed.

- To reduce manual effort and time involved in traditional lost and found processes.

1.5 SCOPE OF THE PROJECT

The *Online Lost and Found Portal* is designed to provide an efficient and centralized platform for managing lost and found items within institutions, organizations, and public environments. The scope of the project includes enabling users to register, log in, and report lost or found items with relevant details such as description, category, date, location, and images.

The system allows users to search and filter items based on different criteria, helping them quickly identify potential matches. It also provides a personalized dashboard where users can view, update, and manage their submitted reports. The platform ensures that all data is stored securely and can be accessed easily whenever required.

CHAPTER 2

LITERATURE SURVEY

2.1 WEB-BASED LOST AND FOUND SYSTEMS

Web-based lost and found systems have been developed to address the limitations of traditional methods used for managing lost items. In many institutions, lost items are recorded using manual registers or displayed on notice boards, which often leads to poor organization and difficulty in retrieving information. To overcome these challenges, researchers have proposed centralized web platforms that allow users to report and search for lost and found items efficiently.

According to various studies, web-based systems provide a structured approach where users can submit detailed information such as item name, description, category, date, and location. These details are stored in a centralized database, making it easier to access and manage the information. Users can search for items using keywords or filters, which significantly reduces the time required to locate lost belongings. Additionally, these systems provide better transparency compared to traditional methods, as all users can view the available records.

Another important advantage of web-based systems is accessibility. Since the platform is available online, users can access it from anywhere at any time. This increases user participation and improves the chances of recovering lost items. Furthermore, modern web technologies enable the development of responsive and user-friendly interfaces, making the system easy to use even for non-technical users.

However, some existing systems lack advanced features such as intelligent matching and real-time notifications. Despite these limitations, web-based lost and found systems represent a significant improvement over manual approaches and form the foundation for more advanced solutions.

2.2 MOBILE AND ONLINE ITEM TRACKING SYSTEMS

With the rapid growth of smartphones and mobile applications, mobile-based lost and found systems have gained popularity in recent years. These systems extend the functionality of web platforms by providing users with the ability to report and search for items directly from their mobile devices. This increases convenience and ensures that users can interact with the system in real time.

Mobile and online tracking systems often include features such as location-based services, which allow users to tag the exact location where an item was lost or found. This helps in narrowing down search results and improves the accuracy of matching items. Users can also upload images instantly, providing visual proof that enhances item identification.

Another key advantage of mobile systems is instant communication. Users can receive notifications when a matching item is found, reducing the delay in recovery. These systems also encourage active user participation, as reporting an item becomes quick and easy.

Despite these benefits, mobile-based systems face certain challenges. They depend heavily on internet connectivity and require users to provide accurate information. In addition, not all users may be willing to install or regularly use such applications. Nevertheless, mobile and online tracking systems play an important role in improving the efficiency and accessibility of lost and found services.

2.3 DATABASE MANAGEMENT IN TRACKING SYSTEMS

Database management is a crucial component of any lost and found system, as it ensures efficient storage, retrieval, and organization of data. In such systems, large amounts of information related to lost and found items are generated ежедневно, including descriptions, images, locations, and timestamps. Without proper database management, handling this data would become difficult and inefficient.

Modern systems use structured databases to store item details in an organized manner. Each record is stored with specific attributes, allowing users to search and filter data easily. Database indexing techniques further improve performance by enabling faster query processing.

This ensures that users can retrieve relevant information quickly, even when the database contains a large number of records.

Another important aspect is data consistency and security. Database systems ensure that information is stored accurately and can be updated when necessary. Access control mechanisms can also be implemented to restrict unauthorized access and protect sensitive user data.

Efficient database design also supports scalability, allowing the system to handle increasing amounts of data as more users join the platform. Overall, database management plays a vital role in ensuring the reliability, performance, and effectiveness of lost and found systems.

2.4 SEARCH AND FILTERING TECHNIQUES

Search and filtering techniques are essential features in lost and found systems, as they help users quickly locate relevant items from a large dataset. Basic search functionality allows users to enter keywords related to the lost item, such as name or description, and retrieve matching results.

Advanced filtering options further improve the search process by allowing users to narrow down results based on specific criteria such as category, location, date, and item type. This reduces the number of irrelevant results and makes the search more efficient.

Modern systems also implement intelligent search techniques that analyze user input and provide more accurate results. For example, partial matching and keyword suggestions can help users find items even if the exact terms are not used. These techniques enhance user experience and reduce the effort required to locate items.

Effective search and filtering mechanisms are important for improving system usability. They ensure that users can quickly find relevant information without browsing through large amounts of data. As a result, these features significantly increase the success rate of recovering lost items.

2.5 ARTIFICIAL INTELLIGENCE IN ITEM MATCHING

Artificial intelligence has become an important technology in improving the functionality of lost and found systems. AI-based techniques can analyze large amounts of data and identify patterns that may not be easily detected by humans. In the context of lost and found systems, AI is used to match lost items with found items based on similarities in descriptions, images, and other attributes.

Machine learning algorithms can be trained to recognize patterns in item data and suggest possible matches to users. For example, if a user reports a lost item with a specific description and color, the system can automatically identify similar items in the database. This reduces manual effort and increases the chances of successful recovery.

AI can also be used for image recognition, where uploaded images are analyzed to identify objects and compare them with other images in the system. This adds an additional layer of accuracy to the matching process.

2.6 CLOUD-BASED WEB APPLICATIONS

Cloud computing has transformed the way web applications are developed and deployed. In lost and found systems, cloud-based platforms provide several advantages, including scalability, accessibility, and data security.

Cloud-based systems allow data to be stored on remote servers, making it accessible from anywhere with an internet connection. This enables users to interact with the system in real time and ensures that updates are reflected instantly. It also allows multiple users to access the platform simultaneously without affecting performance.

Another advantage is scalability. Cloud platforms can handle increasing amounts of data and user traffic without requiring major changes to the system. This makes them suitable for large-scale applications.

Security is also an important aspect of cloud computing. Advanced security measures such as encryption and access control are used to protect user data. However, proper implementation is necessary to prevent unauthorized access.

CHAPTER 3

EXISTING SYSTEM

3.1 EXISTING SYSTEM

The existing system for managing lost and found items in institutions and public places is primarily based on manual methods. These methods include maintaining physical registers, displaying notices on boards, and relying on verbal communication among individuals. While these approaches have been used for a long time, they are not efficient in handling large volumes of data and often lead to confusion and mismanagement.

In a typical existing system, when a person loses an item, they report it to an authority or write the details in a register. Similarly, if someone finds an item, they may submit it to an office or announce it informally. The information is then recorded manually, which makes it difficult to organize and retrieve when needed. There is no centralized platform where all users can view or search for lost and found items easily.

Another major limitation of the existing system is the lack of proper communication between users. A person who has lost an item may not be aware if it has been found by someone else, as there is no direct way to connect both parties. This significantly reduces the chances of recovering lost belongings. Additionally, manual systems do not support features such as image uploads, detailed descriptions, or real-time updates, which are important for accurate identification of items.

The existing system is also time-consuming and prone to errors. Important details may be missed, records can be misplaced, and there is no guarantee that the information is up-to-date. Furthermore, accessing past records is difficult, as it requires manually searching through registers or files.

Overall, the current system lacks efficiency, transparency, and accessibility. These limitations highlight the need for a modern, digital solution that can effectively manage lost and found items and improve the chances of successful recovery.

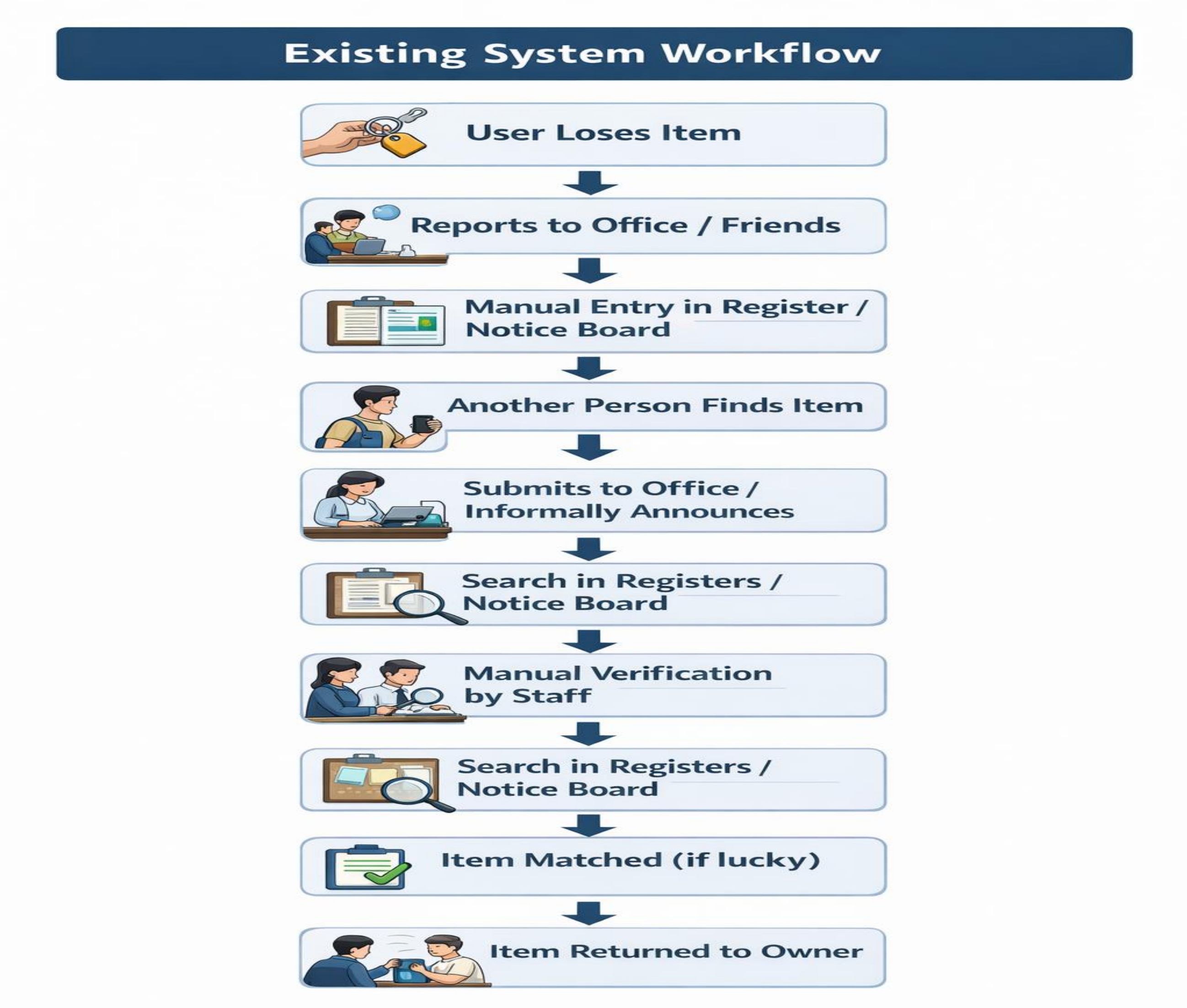

Figure 3.1 Existing System

3.2 DISADVANTAGES

- Lack of spatial understanding due to 2D images and sketches.
- Manual documentation is time-consuming and prone to human error.
- Difficult to accurately reconstruct complex crime scenes.
- Limited visualization makes it hard to analyze evidence clearly.
- High dependency on investigator's experience and interpretation.
- Incomplete or inconsistent data collection may affect investigation results.
- No interactive or 360-degree view of the crime scene.
- Difficulty in managing and organizing large amounts of forensic data.
- Risk of data loss or damage due to lack of secure storage systems.
- Limited collaboration between investigators and forensic experts.

CHAPTER 4

PROBLEMS IDENTIFIED

The existing system for managing lost and found items suffers from several limitations that reduce its efficiency and reliability. These problems arise mainly due to the use of manual processes and the absence of a centralized digital platform. Identifying these issues is important in order to develop an improved and effective solution.

One of the major problems is the lack of a centralized system. Information about lost and found items is usually scattered across different sources such as registers, notice boards, or informal communication. This makes it difficult for users to access complete and accurate information at a single place.

Another significant issue is the time-consuming nature of the process. Users must manually report items and search through records, which requires considerable effort and time. This delay reduces the chances of recovering lost items quickly. Additionally, there is no real-time update mechanism, so users may not be aware when a matching item is found.

The existing system also lacks proper communication between users. There is no direct link between the person who lost an item and the person who found it. As a result, even if an item is found, it may not reach the rightful owner due to communication gaps.

Data management is another major concern. Manual records are prone to errors, duplication, and loss of information. Important details may be incomplete or incorrectly recorded, making it difficult to identify items accurately. Moreover, maintaining and searching through physical records becomes increasingly difficult as the volume of data grows.

The absence of advanced features such as image uploads, search filters, and intelligent matching further limits the effectiveness of the system. Without visual proof or proper categorization, identifying items becomes challenging.

CHAPTER 5

PROPOSED SYSTEM

5.1 PROPOSED SYSTEM OVERVIEW

The proposed system, *Online Lost and Found Portal*, is a web-based application designed to overcome the limitations of the existing manual system by providing a centralized, efficient, and user-friendly platform for managing lost and found items. The system aims to digitize the entire process, allowing users to easily report, search, and recover lost belongings in a structured manner.

In the proposed system, users are required to register and log in to access the platform. Once authenticated, users can report lost or found items by providing detailed information such as item title, description, category, color, location, date, and images. This information is stored in a centralized database, making it easily accessible for searching and matching purposes

5.2 IMAGE ACQUISITION

In the *Online Lost and Found Portal*, image acquisition refers to the process of capturing and uploading images of lost or found items into the system. Images play a crucial role in improving the identification and matching of items, as visual information provides more clarity compared to text descriptions alone.

The system allows users to upload images while reporting a lost or found item. These images can be captured using mobile devices or cameras and then uploaded through the web interface. The platform supports common image formats such as JPEG and PNG, ensuring compatibility and ease of use for all users.

Once uploaded, the images are stored securely in the system and linked with the corresponding item details such as title, description, location, and date. These images serve as visual proof and help other users quickly recognize and verify items. This significantly increases the chances of successful matching and recovery.

5.3 AI-BASED ITEM MATCHING

In the *Online Lost and Found Portal*, AI-based item matching is used to improve the process of identifying and connecting lost items with found items. Instead of manually comparing each entry, the system utilizes intelligent techniques to analyze item details and suggest possible matches automatically.

The matching process is based on various attributes such as item title, description, category, color, location, and uploaded images. When a user reports a lost or found item, the system compares the provided data with existing records in the database to identify similarities. This reduces the time and effort required for manual searching.

In addition to text-based matching, the system can incorporate AI or API-based assistance to enhance accuracy. For example, image analysis and pattern recognition techniques can be used to compare visual features of items. This helps in identifying matches even when the descriptions are not exactly the same.

The system may also provide suggestions or recommendations to users by displaying similar items in the dashboard or search results. This improves user experience and increases the chances of successful recovery.

5.4 VISUALIZATION

In the *Online Lost and Found Portal*, visualization refers to the graphical representation and display of lost and found item data in a clear, structured, and user-friendly manner. Effective visualization plays an important role in helping users quickly understand information, identify items, and interact with the system efficiently.

The system provides an intuitive user interface where all items are displayed in the form of organized cards or lists. Each item includes essential details such as title, description, category, date, location, and an image. The use of images significantly enhances visualization by allowing users to visually verify items instead of relying only on textual descriptions.

5.5 PROPOSED SYSTEM ARCHITECTURE

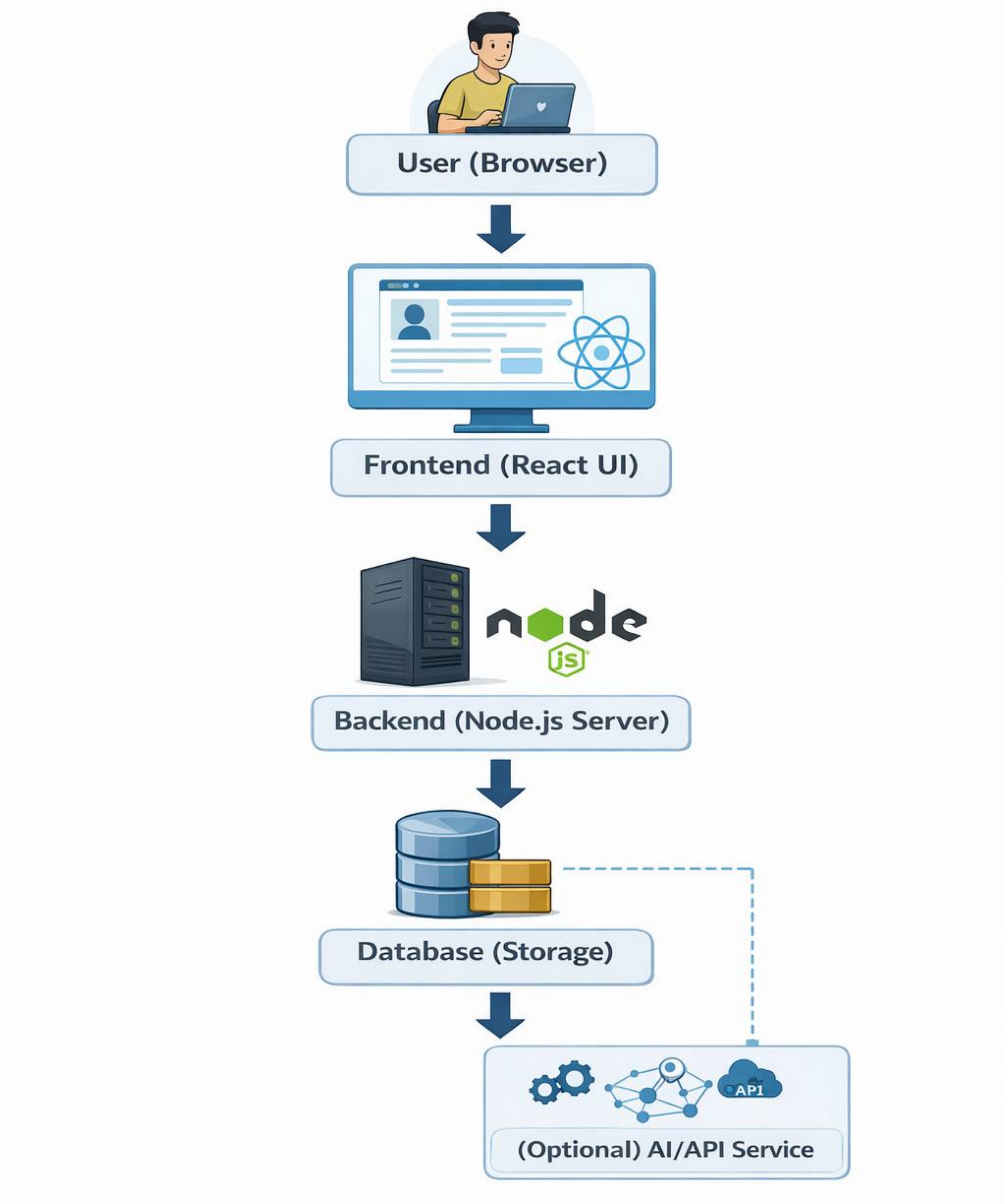

Figure 5.5 Proposed System

The overall workflow of the proposed system begins with users accessing the portal through a web browser and logging into their accounts. Users can then report lost or found items by entering relevant details such as title, description, category, location, date, and uploading images. The submitted data is validated and stored in a centralized database. The system then processes this information and enables search and filtering features for other users.

Additionally, AI-based matching techniques may be used to identify similar items and suggest possible matches. The results are displayed through an interactive user interface, allowing users to easily browse and verify items. All data is securely stored and managed within the system, ensuring easy retrieval and efficient tracking of lost and found items.

5.6 ADVANTAGES

- Provides a **centralized platform** to manage all lost and found items in one place.
- Enables **quick reporting and searching**, saving time compared to manual methods.
- Supports **image uploads**, improving accuracy in identifying items.
- Offers **advanced search and filtering** for easy item discovery.
- Ensures **secure data management** with user authentication.
- Includes **AI-based matching** to suggest possible item matches automatically.
- Provides a **user-friendly and responsive interface** accessible on multiple devices.

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

• **Processor**

A computer system with a modern processor such as Intel Core i3/i5 or higher is required to run the application efficiently. The processor handles frontend rendering, backend execution, and overall system performance.

• **Memory (RAM)**

A minimum of 4 GB RAM is required, while 8 GB is recommended for smooth performance. Adequate memory ensures efficient multitasking, faster loading of the application, and smooth execution of server-side operations.

• **Storage**

At least 256 GB of storage (preferably SSD) is required to store project files, application data, and database records. SSD storage improves speed and system responsiveness.

• **Input Device**

A keyboard and mouse are required for system interaction. A smartphone or camera can be used to capture images of lost or found items for uploading into the system.

• **Internet Connectivity**

A stable internet connection is required for accessing the web application, uploading images, and enabling communication between frontend and backend services.

6.2 SOFTWARE REQUIREMENTS

• **Operating System**

The system can be developed and executed on Windows, Linux, or macOS. These operating systems support web development tools and server environments.

- **Frontend (React + Vite)**

  React is used to build the user interface of the system, providing a dynamic and responsive experience. Vite is used as the development tool for faster build and performance.

- **Backend (Node.js)**

  Node.js is used for server-side development. It handles user requests, authentication, item management, and communication with the database.

- **Framework / API**

  REST APIs are used for communication between frontend and backend. These APIs handle data exchange, user actions, and system operations.

- **Database**

  A database such as MySQL, MongoDB, or JSON-based storage is used to store user data and item details securely.

- **Development Environment**

  Visual Studio Code (VS Code) is used for coding, debugging, and managing project files efficiently.

- **Web Browser**

  Modern browsers such as Google Chrome, Edge, or Firefox are required to run and test the application.

CHAPTER 7

SYSTEM IMPLEMENTATIONS

7.1 LIST OF MODULES

The *Online Lost and Found Portal* is divided into several functional modules to ensure efficient operation and better organization of the system. Each module is responsible for handling a specific task within the application. The main modules of the system are as follows:

- User Authentication Module
- Item Reporting Module (Lost & Found)
- Search and Filtering Module
- AI-Based Matching Module
- User Dashboard Module
- Notification Module
- Database Management Module

7.2 MODULE DESCRIPTION

7.2.1 User Authentication Module

This module handles user registration and login functionality. It allows new users to create an account and existing users to log in securely. The system verifies user credentials and ensures that only authorized users can access the platform. This module also manages session handling and logout functionality, providing security and controlled access to the system.

7.2.2 ITEM-REPORTING

The item reporting module enables users to report lost or found items by entering relevant details such as title, description, category, location, date, and uploading images. This module ensures that all item data is properly validated and stored in the database. It forms the core functionality of the system by allowing users to share information about items.

7.2.3 Search and Filtering Module

This module allows users to search for items using keywords and apply filters such as category, location, and date. It helps users quickly find relevant items from the database. The search functionality improves user experience by reducing the effort required to locate lost belongings.

7.2.4 AI-Based Matching Module

The AI-based matching module analyzes item details and identifies similarities between lost and found items. It uses attributes such as description, category, and images to suggest possible matches. This module reduces manual effort and increases the chances of successfully recovering items.

7.2.5 User Dashboard Module

The dashboard provides a personalized interface where users can view and manage their reported items. Users can track lost and found items, update information, or delete posts if necessary. It gives a clear overview of user activity within the system.

CHAPTER 8
SYSTEM TESTING

8.1 UNIT TESTING

Unit testing is the process of testing individual components or modules of a system independently to ensure that each part functions correctly. In the *Online Lost and Found Portal*, unit testing is performed on each module such as user authentication, item reporting, search functionality, and database operations.

During unit testing, each module is tested with different input values to verify its correctness. For example, in the user authentication module, test cases are created to check whether users can successfully register and log in with valid credentials, and whether invalid inputs are properly handled. Similarly, the item reporting module is tested to ensure that all fields such as title, description, location, and image upload work correctly and data is stored accurately.

8.2 INTEGRATION TESTING

Integration testing is performed to verify that different modules of the system work together correctly as a complete system. After successful unit testing of individual components, the modules of the *Online Lost and Found Portal* are combined and tested to ensure proper interaction and data flow between them.

In this system, integration testing focuses on the interaction between modules such as user authentication, item reporting, search and filtering, dashboard, and database management. For example, when a user logs in through the authentication module, the system should correctly grant access to the dashboard and allow the user to report or search for items. Similarly, when a user submits a lost or found item, the data should be successfully passed to the database and reflected in the search results.

8.3 SYSTEM TESTING

System testing is the process of evaluating the complete and integrated system to ensure that it meets the specified requirements and functions correctly in a real-world environment.

During system testing, the entire application is tested as a whole, including all modules such as user authentication, item reporting, search and filtering, AI-based matching, dashboard, and database management. The system is tested with different scenarios to ensure that all features work correctly under various conditions.

Non-functional aspects such as performance, usability, and security are also considered during system testing. The application is tested to ensure it loads properly, responds quickly to user actions, and provides a smooth user experience. Security checks are performed to verify that user data is protected and unauthorized access is prevented.

8.4 PERFORMANCE TESTING

Performance testing is conducted to evaluate how well the system performs under different conditions, including varying workloads and user activities. In the *Online Lost and Found Portal*, performance testing ensures that the application responds quickly, handles multiple users efficiently, and maintains stability during operation.

8.5 USABILITY TESTING

Usability testing is performed to evaluate how easily users can interact with the system and how effectively they can complete tasks using the application. In the *Online Lost and Found Portal*, usability testing focuses on ensuring that the interface is user-friendly, intuitive, and accessible to all users, including those with minimal technical knowledge.

During usability testing, users are asked to perform common tasks such as registering an account, logging in, reporting a lost or found item, uploading images, searching for items, and navigating the dashboard.

Their interactions with the system are observed to identify any difficulties or confusion they may encounter while using the application.

The design and layout of the interface are evaluated to ensure clarity and simplicity. Elements such as buttons, menus, forms, and navigation links are tested to verify that they are easy to understand and use.

CHAPTER 9
RESULT AND DISCUSSION

The performance comparison between the existing system and the proposed *Online Lost and Found Portal* clearly demonstrates the effectiveness of the proposed approach in improving the management and recovery of lost items. The evaluation is based on key factors such as efficiency, response time, data management, user experience, and accuracy in item matching. The existing system relies on manual methods such as registers and notice boards, which result in poor organization and reduced chances of recovering lost items. In contrast, the proposed system provides a centralized digital platform that significantly improves efficiency and accessibility.

In terms of processing speed, the existing system is time-consuming as users must manually report and search for items. The proposed system automates these processes by allowing users to quickly submit item details and perform searches using filters and keywords. This reduces the time required to find relevant information and improves overall system performance. Additionally, the use of a web-based interface enhances user experience by providing a simple and interactive platform for managing lost and found items.

Data management is also greatly improved in the proposed system. Unlike manual records that are prone to loss and errors, the system stores all information securely in a structured database. This ensures accurate storage, easy retrieval, and efficient handling of large amounts of data. The inclusion of image uploads further enhances the accuracy of item identification, making it easier for users to verify items.

CHAPTER 10

CONCLUSION AND FUTURE ENHANCEMENTS

10.1 CONCLUSION

The *Online Lost and Found Portal* was developed to improve the process of managing and recovering lost items using modern web technologies. Traditional methods such as manual registers, notice boards, and verbal communication are inefficient and often fail to connect lost items with their rightful owners. These limitations make the process time-consuming and unreliable.

The proposed system addresses these challenges by providing a centralized digital platform where users can report, search, and manage lost and found items بـ سهولة. The system integrates multiple modules including user authentication, item reporting, search and filtering, AI-based matching, dashboard management, and database storage. These modules work together to ensure smooth operation and efficient handling of data.

The implementation and testing results show that the system performs effectively. Users can easily register, log in, upload item details with images, and search for relevant items. The search and filtering features improve accessibility, while the structured database ensures accurate data management. The system also enhances user experience through a simple and interactive interface.

Overall, the *Online Lost and Found Portal* demonstrates how digital solutions can significantly improve efficiency, reduce manual effort, and increase the chances of recovering lost items. It provides a reliable, scalable, and user-friendly solution suitable for institutions and public environments.

10.2 FUTURE ENHANCEMENTS

Although the proposed system performs effectively, several enhancements can be implemented in the future to improve its functionality and performance. One possible enhancement is the integration of advanced artificial intelligence techniques for more accurate item matching. Another improvement is the addition of a real-time notification system, such as email or mobile alerts, to inform users immediately when a matching item is found.

The system can also be extended by developing a mobile application, which would increase accessibility and allow users to report items instantly using their smartphones.

Integration with GPS or location-based services can also be added to improve tracking and help users identify where items were lost or found more accurately.

Another enhancement is the use of cloud-based storage to improve scalability, data security, and accessibility. This would allow the system to handle a larger number of users and data efficiently.

Additionally, features such as admin control, verification mechanisms, and chat functionality between users can be implemented to improve communication and ensure authenticity of data.

By implementing these enhancements, the system can become more intelligent, efficient, and widely usable, making it a powerful solution for managing lost and found items in modern environments.

APPENDIX – A

SOURCE CODE

1. Home Page (App.tsx)

```
import { BrowserRouter as Router, Routes, Route } from "react-router-dom";
import Login from "./pages/Login";
import Feed from "./pages/Feed";
import PostItem from "./pages/PostItem";
import Dashboard from "./pages/Dashboard";
import ItemDetail from "./pages/ItemDetail";

function App() {
  return (
    <Router>
      <Routes>
        <Route path="/" element={<Feed />} />
        <Route path="/login" element={<Login />} />
        <Route path="/post" element={<PostItem />} />
        <Route path="/dashboard" element={<Dashboard />} />
        <Route path="/item/:id" element={<ItemDetail />} />
      </Routes>
    </Router>
  );
}
export default App;
```

2. Login Page (Login.tsx)

```
import React, { useState } from "react";

export default function Login() {
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");

  const handleLogin = () => {
```

```
    if (email && password) {
      alert("Login successful");
    } else {
      alert("Enter valid details");
    }
  };
  return (
    <div>
      <h2>Login</h2>
      <input
        type="email"
        placeholder="Enter email"
        onChange={(e) => setEmail(e.target.value)}
      />
      <input
        type="password"
        placeholder="Enter password"
        onChange={(e) => setPassword(e.target.value)}
      />
      <button onClick={handleLogin}>Login</button>
    </div>
  );
}
```

3. Post Item Page (PostItem.tsx)

```
import React, { useState } from "react";

export default function PostItem() {
  const [title, setTitle] = useState("");
  const [description, setDescription] = useState("");

  const handleSubmit = () => {
    alert("Item Posted Successfully");
```

```
  };

      <textarea
        placeholder="Description"
        onChange={(e) => setDescription(e.target.value)}
      />

      <button onClick={handleSubmit}>Submit</button>
    </div>
  );
}
```

4. Backend (server.ts)

```
import express from "express";
import cors from "cors";

const app = express();
app.use(cors());
app.use(express.json());

let items: any[] = [];

app.post("/add-item", (req, res) => {
  const item = req.body;
  items.push(item);
  res.json({ message: "Item added successfully" });
});

app.listen(5000, () => {
  console.log("Server running on port 5000");
});
```

APPENDIX – B

SCREENSHOTS

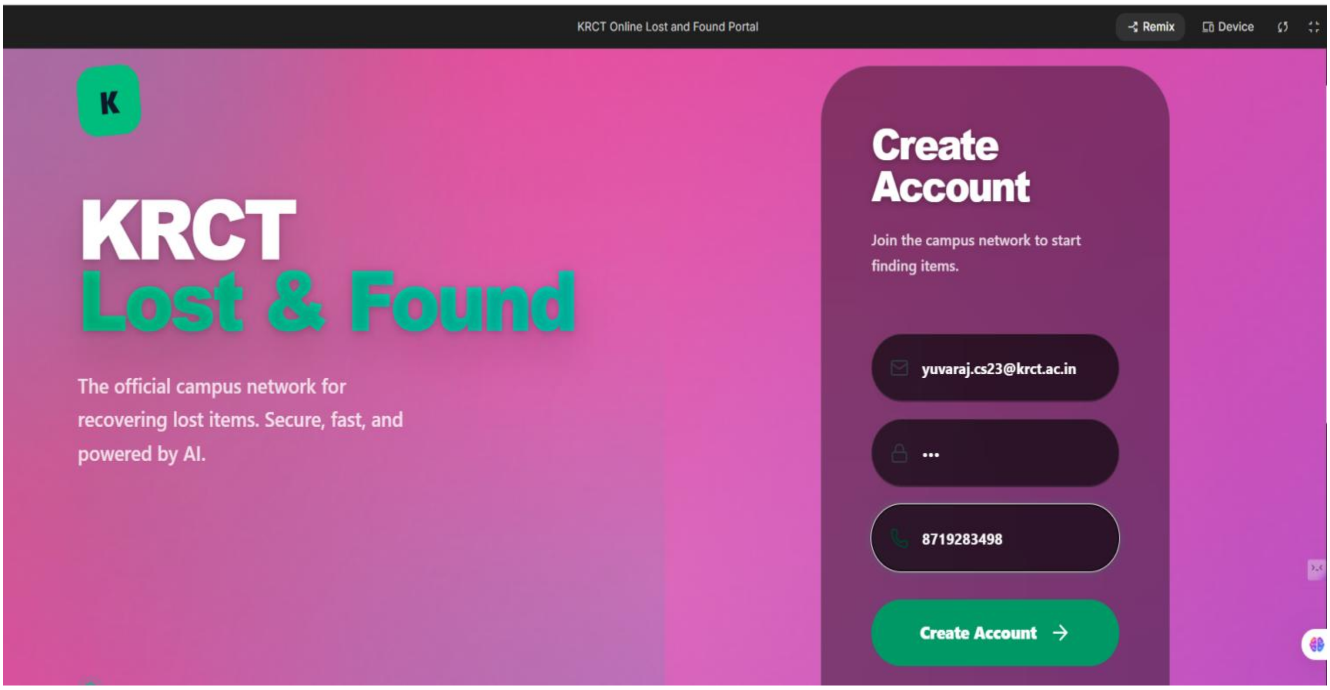

Fig B.1 - Signup Page

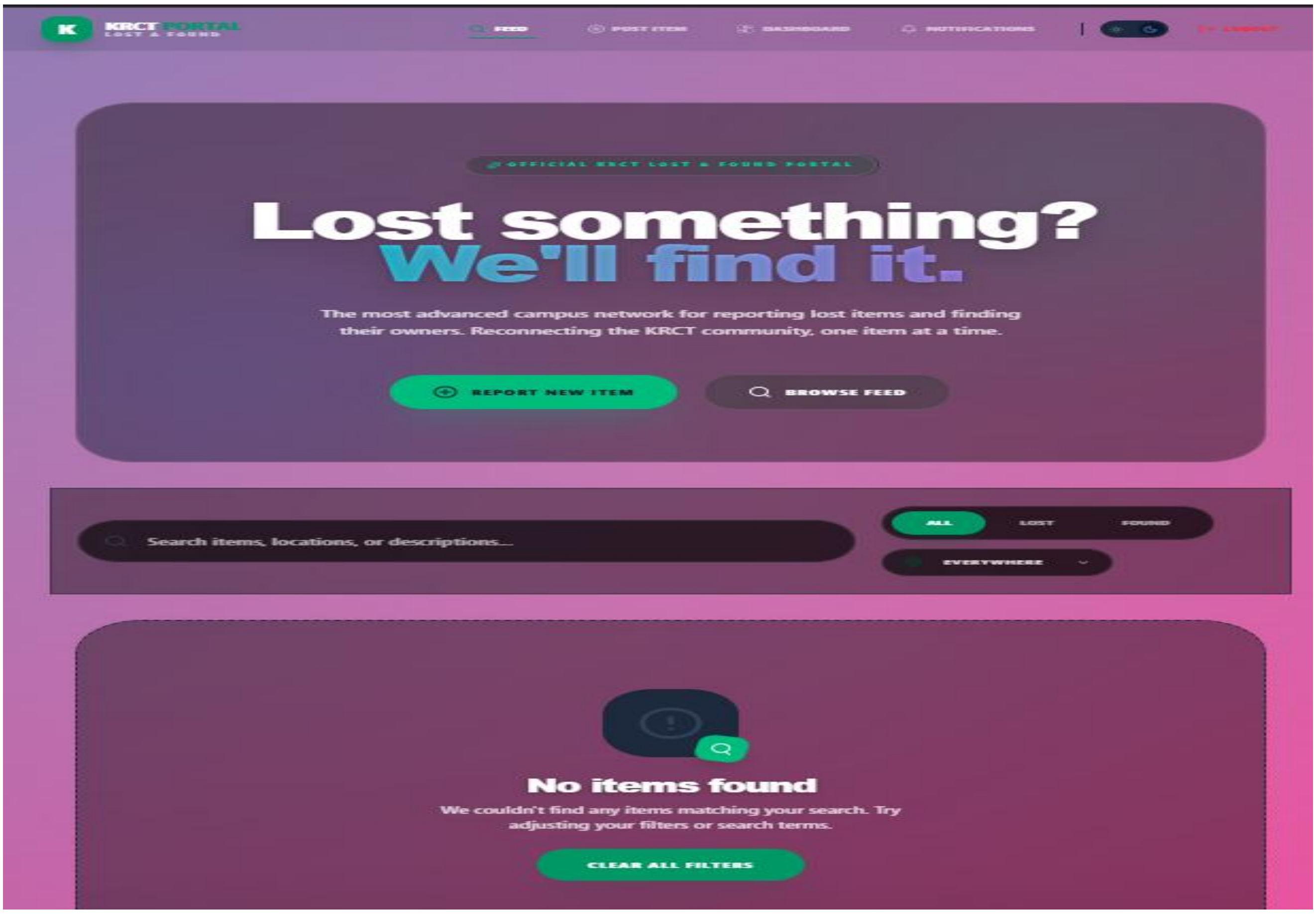

Fig B.2 - Home Page

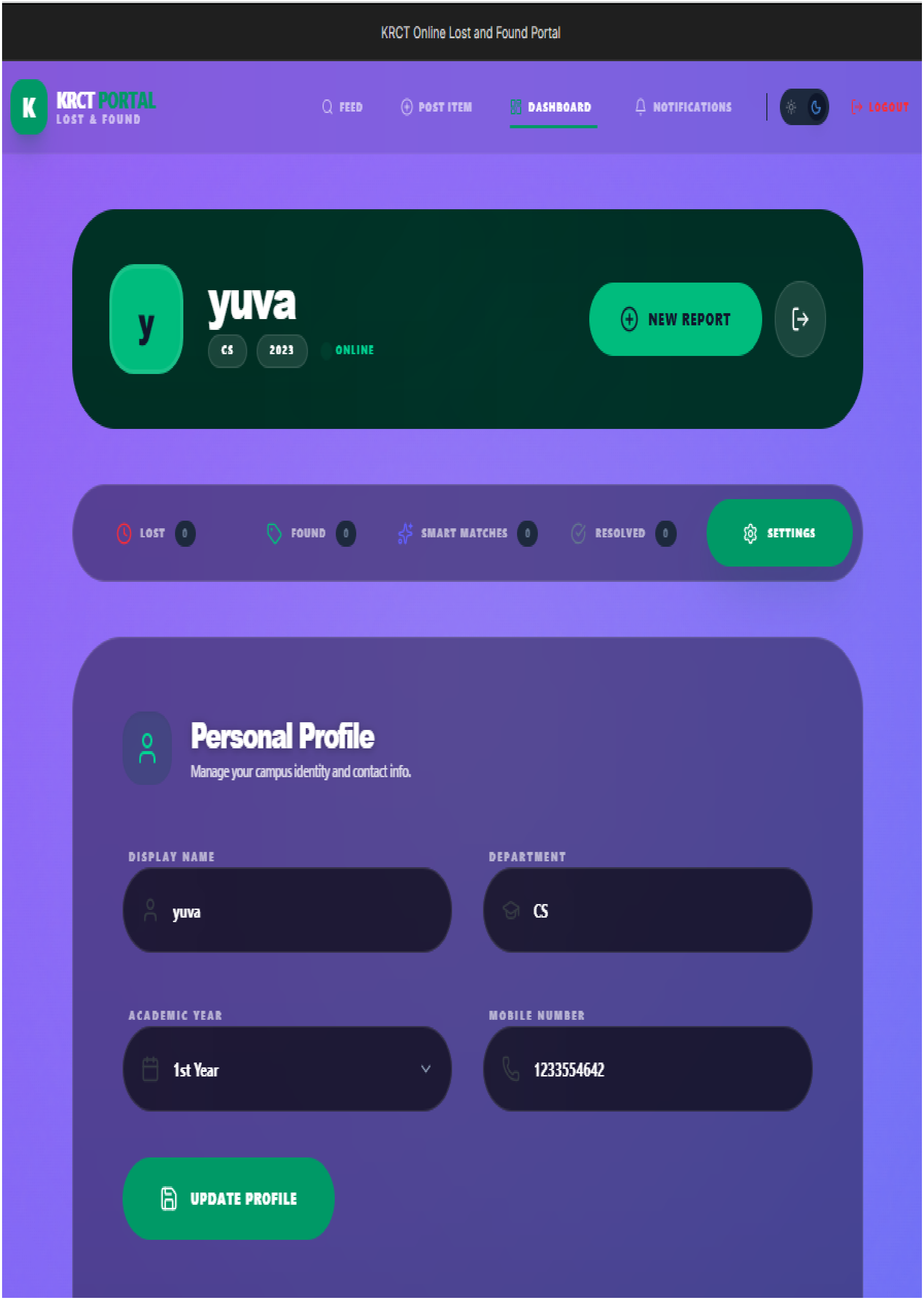

Fig B.3 - User Profile Page

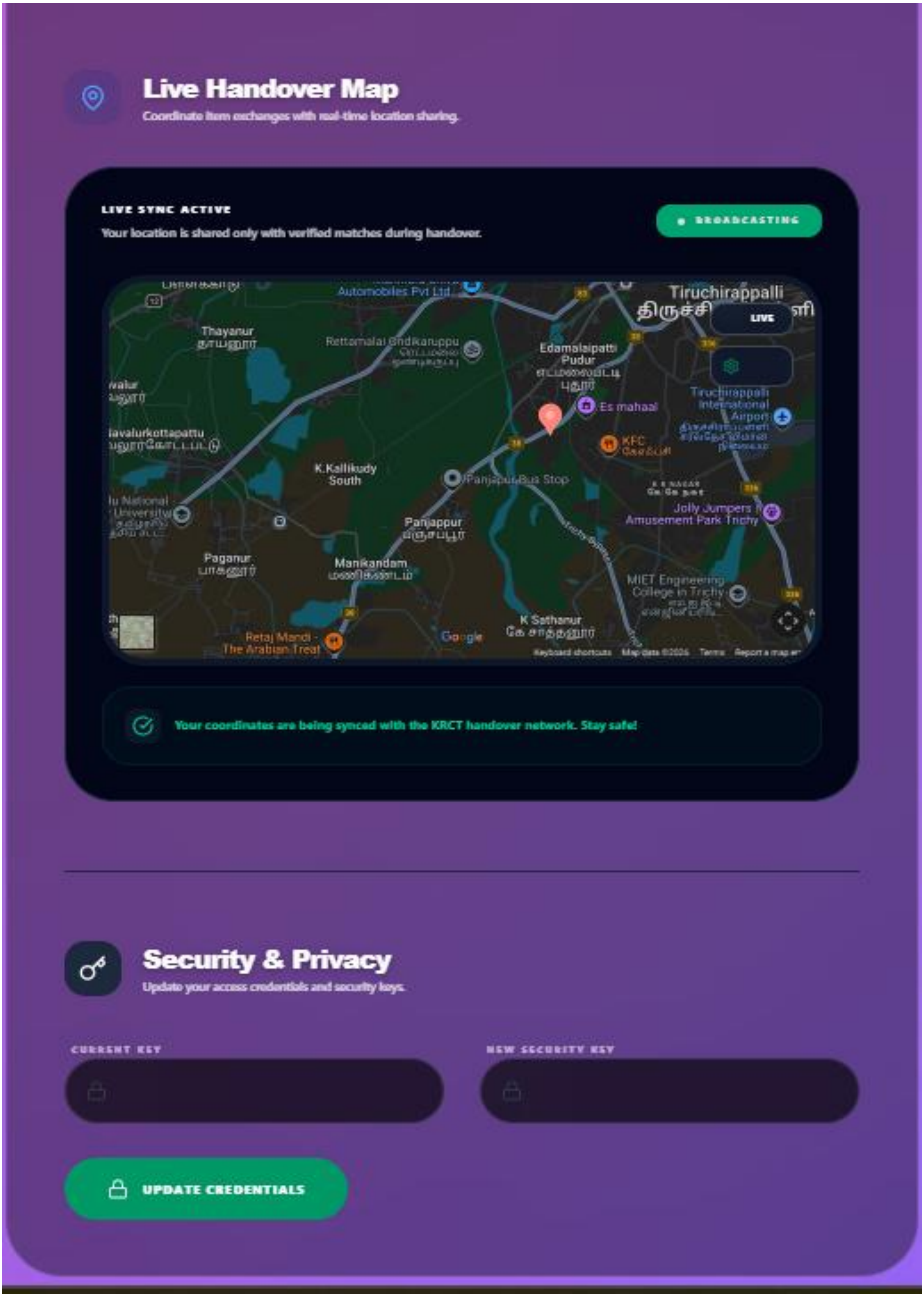

Fig B.4 - Location Page

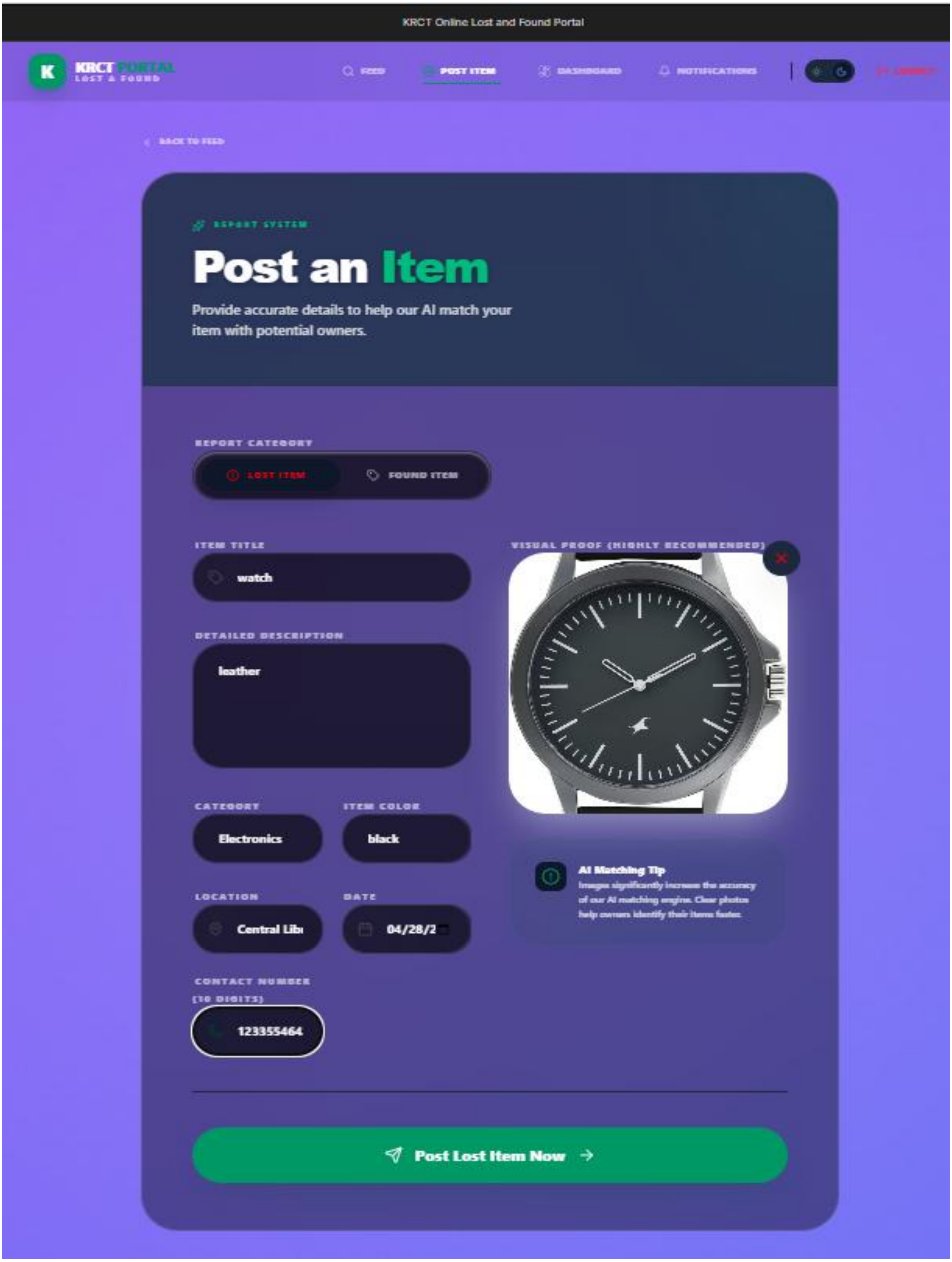

Fig B.5 - Lost Item Post Page

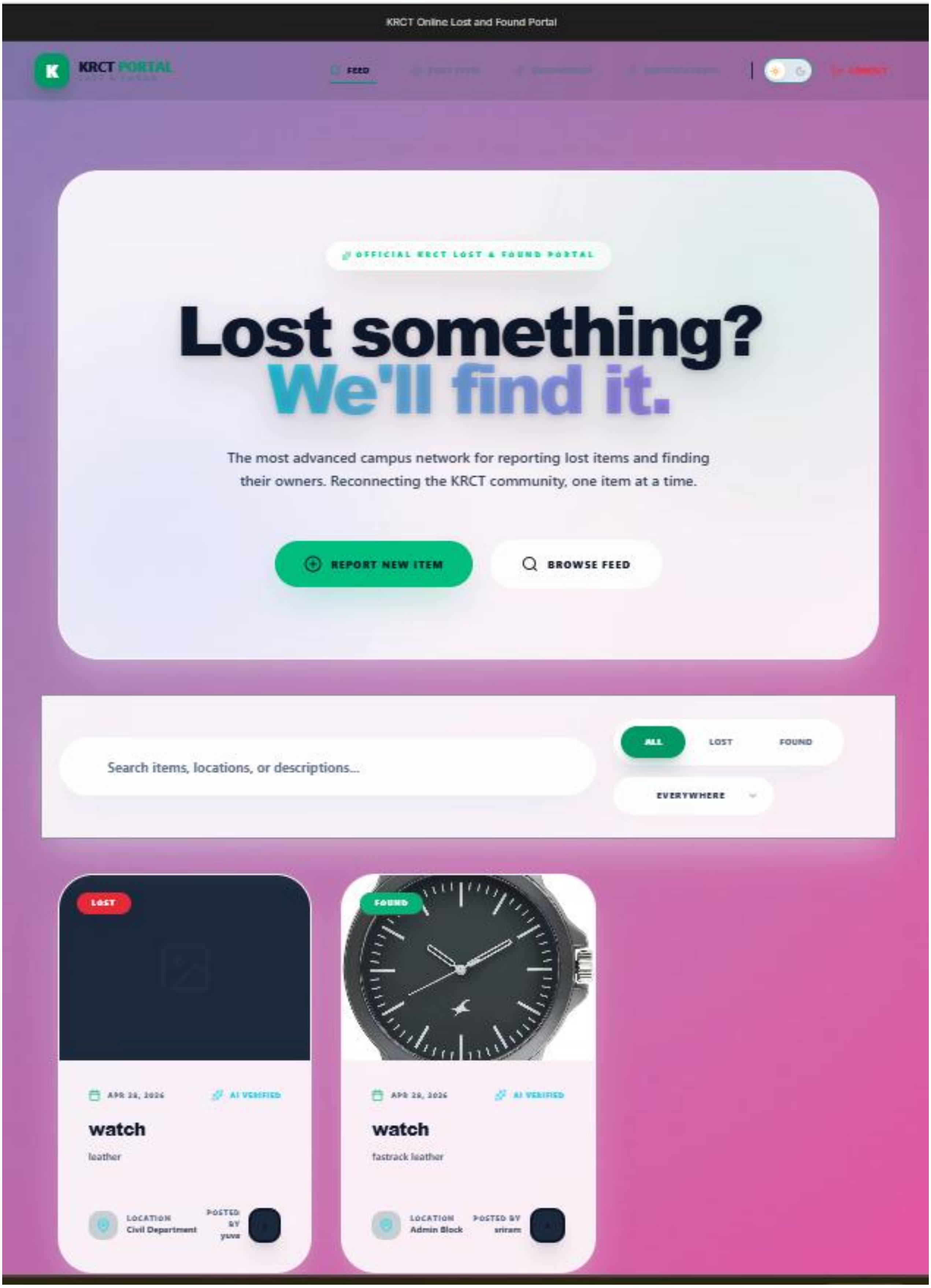

Fig B.6 - Profile Feed

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